Life Cycle Assessment of Turf vs. Traditional Lawns

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1. Introduction

Literature Review

Green grass or traditional lawns in the US typically consist of non-native grass species, like Kentucky Bluegrass, which often have extremely high irrigation and fertilizer needs. Traditional lawns can “account for 75% of total household water consumption in arid regions of the USA” and contribute to increased lawn chemical and fertilizer use (Smetana and Crittenden, 2014). A life cycle assessment was conducted in 2014, focusing on traditional lawns in urban areas, finding high levels of environmental impacts in water use, chemical use, and additional economic impacts when compared to native and “sustainable plants” (Smetana and Crittenden, 2014). In addition to a high environmental impact, these traditional lawns often are time-consuming for homeowners, with high maintenance needs. Due to this, artificial turf, or synthetic turf, was designed as an alternative.

In 2022, a life cycle assessment was conducted to compare the environmental footprints of three different artificial turf styles versus a natural grass pitch for a soccer field (Russo, et al., 2022). The findings showed that due to this field’s use as a football playground, the environmental impact is linked to the intensity of use: “for high intensity of use, artificial turf is more environmentally sustainable, and for less intensive use, a form of natural grass is” (Russo, et al., 2022). It is noted that “when the analysis approach is shifted to the entire life cycle, the results appear ambiguous” (Russo, et al., 2022). An additional life cycle assessment comparing artificial and natural turf systems was developed in 2025, within Nordic climates and regions (Säberg, et al., 2025). This study focused on construction and maintenance but expanded to the end-of-life processes as well, looking at eventual transport and disposal of artificial turf as well (Säberg, et al., 2025). Due to the construction and disposal stages, artificial turf was shown to have the highest environmental impacts, but during the maintenance stage, the “natural turf showed the greatest impacts due to diesel consumption and fertilizer application” (Säberg, et al., 2025). Assessing different construction, maintenance, and disposal techniques could significantly reduce environmental impacts from both turf fields (Säberg, et al., 2025).

Rather than comparing non-native grass species to native grass species, non-native grass species will be compared to artificial turf in this life cycle of assessment. LCAs that include non-native grass species on personal lawns will be helpful in comparison with other LCAs that compare natural grass to artificial grass on playing fields. With a focus on personal lawns rather than playing fields, looking at the production stage for a smaller-scale lawn could show a difference from these study results.

Goal Statement

This study will compare artificial turf to natural grass in the context of private lawns to help homeowners explore the most water and energy efficient option for personal and private lawns. These results will be released publicly to provide homeowners with the background data informing their decisions.

- Product system: Natural grass and artificial turf used in private lawns
- System boundary: Cradle to gate analysis for both natural grass and artificial turf with the system boundaries determined by Ecoinvent version 3.12 APOS Unit tools.
- Inputs and outputs: This study will look at the inputs of raw materials, energy, water usage, and fertilizer. The outputs will focus on greenhouse gas emissions and water pollution.
- LCIA method: This study will use the ReCiPe 2016 v1.03, midpoint (I) method to conduct an LCIA to compare different environmental impacts up to the midpoint of the product life focusing on rest of world (RoW) parameters and data providers. The data is assessed with 20-year impact results.

Product Function and Functional Unit

For the purposes of this LCA, the primary function of a lawn will be to provide a safe, cushioned space for recreation. It is important to note that lawns provide a variety of other functions for human and ecosystem health. Having a lawn can decrease soil erosion, regulate air temperature, and reduce noise pollution. It also allows for water drainage, unlike concrete or pavement. Additionally, residential green spaces are proven to have beneficial mental health effects by

reducing stress and improving focus. They are a natural, relaxing environment that can offer privacy along with outdoor leisure (*Environmental Benefits of Healthy Lawns*). Healthy lawns also tend to improve curb appeal, significantly impacting a property's value and the neighborhood's desirability. Historically, lawns have served as a symbol of wealth, indicating that the homeowner had enough resources to maintain a plot of land for non-agricultural purposes.

The functional unit for this study is defined as **one 15,000 square foot lawn**. This functional unit is based on the average lawn size across the area of interest (see "Data Quality Requirements" for geographical data limitations). The average lawn size was calculated by taking the mean of the median average lawn size for each state based on "The Average Yard Size by State and City - Today's Homeowner (2026)" dataset. This calculation was exactly 14,894 square feet, which was then rounded up to 15,000 square feet for simplification. In order for 15,000 square feet of artificial turf to meet this functional unit, the turf must have a medium pile height (between 1.25 to 1.75 inches) and a low to medium density between 12,000 to 16,000 stitches per square meter (What is the ideal thickness for artificial grass?, n.d.). Natural grass (sod) must be between 1 to 1.5 inches thick to meet this functional unit.

Process Flow Diagrams

This study will focus on the cradle-to-gate environmental impacts of artificial turf versus natural grass. Inputs for artificial turf include raw materials: petroleum, natural gas, recycled vulcanized automobile tires, urethane foam, and gravel; energy, fuel, and water usage. Outputs for artificial turf include greenhouse gas emissions and water pollution. Input flows for natural grass include energy, water, fertilizer, and fuel usage. Output flows for natural grass include water pollution, waste, and air emissions. These input and output flows will be discussed regarding the LCIA results.

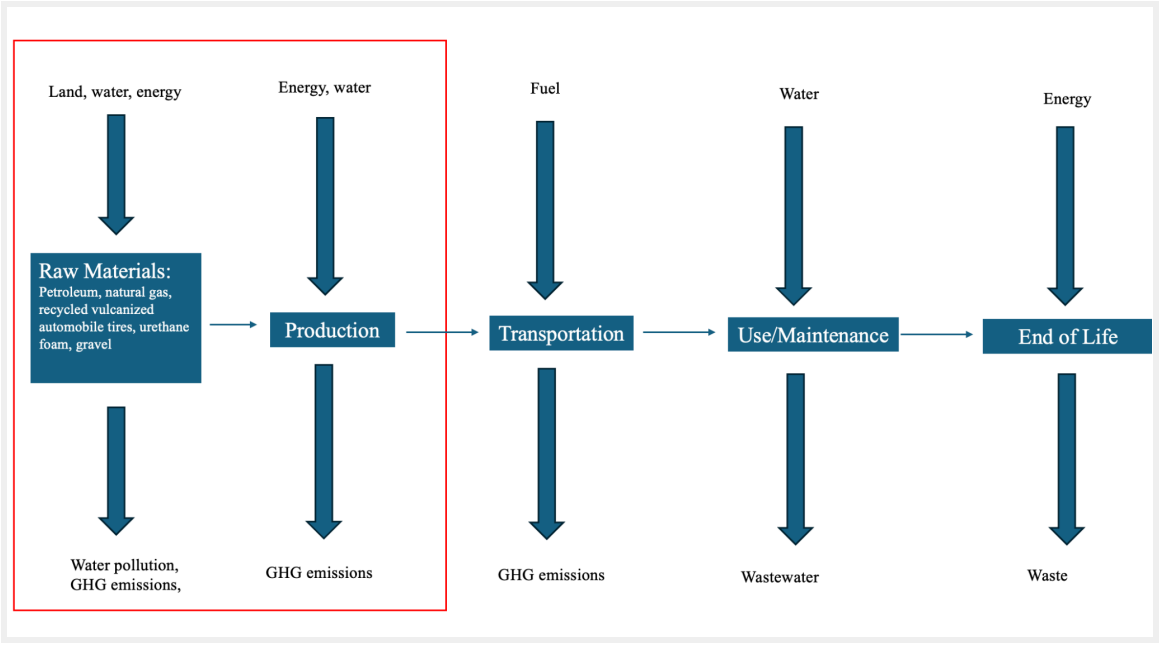


Figure 1a. Process flow diagram for artificial turf.

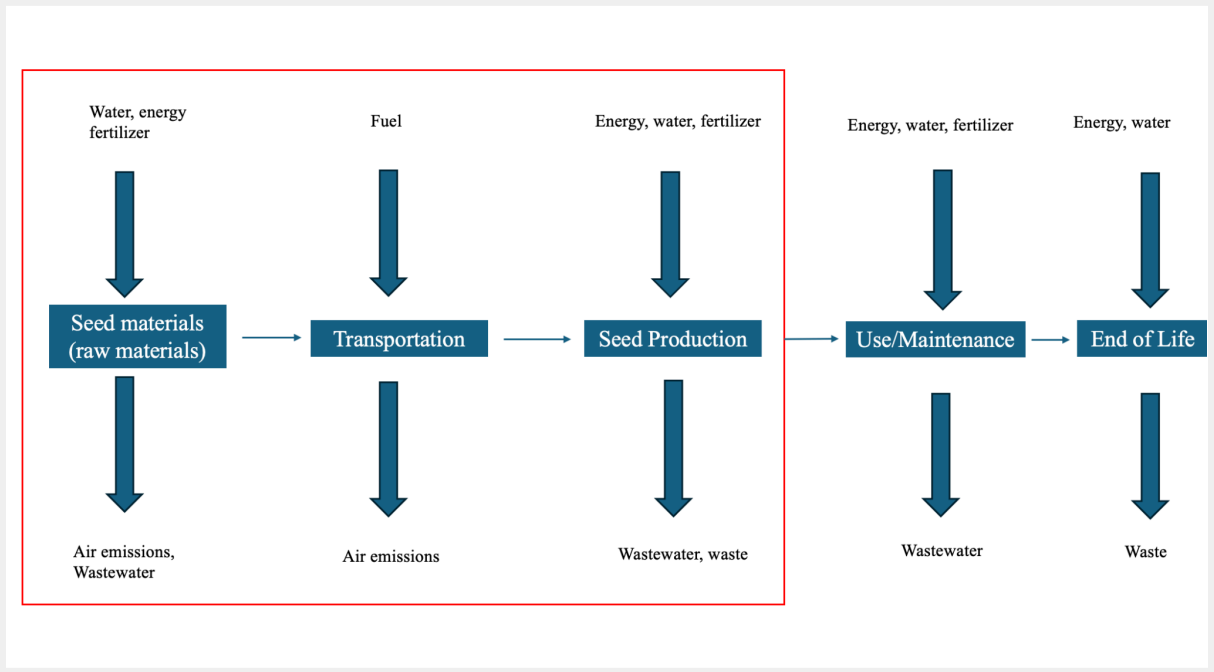


Figure 1b. Process flow diagram for Kentucky Bluegrass (natural grass).

2. Materials and Methods

Life Cycle Impact Assessment (LCIA) Methods

This study uses a comparative LCIA methodology as per ISO 14040 guidelines to evaluate cradle-to-gate environmental impacts of artificial turf versus natural grass. The following steps were taken to conduct this assessment:

1. Define Study Design Parameters
2. Conduct Inventory
3. Conduct Impact Assessment
4. Interpretation of Results

Goal and scope definitions were created to communicate the study's intended application, audience, and public sharing status. Next, product systems were created in OpenLCA v2.6 for both artificial turf and natural grass using the Ecoinvent V 3.2 database. An existing natural grass product system was utilized for the analysis of a natural grass lawn. The artificial turf product system was made from two input processes, carpet production and infill production.

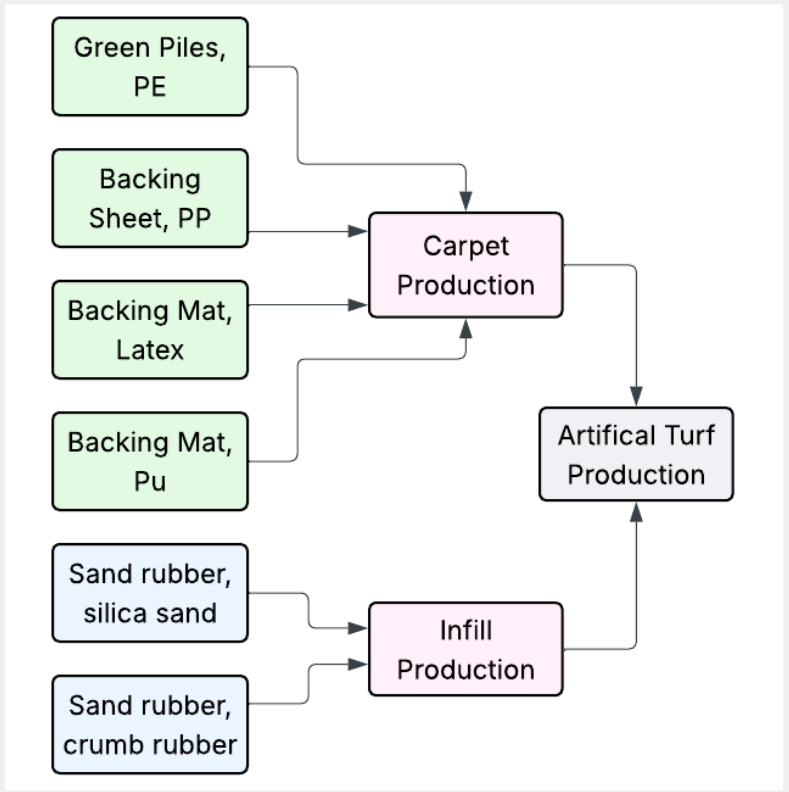


Figure 2. Artificial turf production workflow for OpenLCA analysis.

Flow quantities were calculated by scaling the input quantities from a 2025 study by Falsafi et al. to our functional unit. The inputs (outlined in table 1) were assigned an appropriate pre-existing Ecoinvent flow. To assess relevant environmental impact categories, ReCiPe 2016 v1.03, midpoint (I) impact assessment method was applied, focusing on short-term impacts with lower uncertainty levels.

Table 1. Artificial turf product system inputs. Inputs referenced from Falsafi et al. 2025. Ecoinvent flows are from the Ecoinvent database v3.2 (Wernet et al., 2016).

Inputs	**Value (kg)**	**Ecoinvent Flow**
Carpet Production		
green piles, PE	1834.9	polyethylene, linear low density, granulate
Backing sheet, PP	273.2	Polypropylene, granulate
Backing mat, latex	693	synthetic rubber
Backing mat, Pu	693	polyurethane, flexible foam
Infill Production		
Sand rubber, silica sand	22584.6	silica sand
Sand rubber, crumb rubber	15049.9	used tyre
Total	41128.6	

Table 2. Grass product system inputs (Wernet et al., 2016).

Inputs	**Value (kg)**	**Ecoinvent Flow**
Grass	1602.579	grass, organic

To determine overall model uncertainty, Monte Carlo simulations were used alongside a pedigree matrix. A score for each of the 5 quality indicators– reliability, completeness, temporal correlation, geographical correlation, and further technical correlation, were assigned to both product systems. The scores were informed based on available metadata regarding how, where, and why the data was collected. The Monte Carlo simulation ran the ReCiPe 2016 v1.3 midpoint (I) model 500 times with randomly generated inputs. This created a probability distribution

showing the range of potential outcome values. The mean value in each distribution was then recorded to compare to the non-adjusted impact analysis results.

Table 3. Pedigree matrix input values.

Product	**Reliability**	**Completeness**	**Temporal Correlation**	**Geographical Correlation**	**Further Technological Correlation**
Grass	1	2	1	2	1
Artificial Turf	2	1	1	1	2

Data Quality Requirements

Parameters to ensure data quality have been established for this LCA. All data will meet the Ecoinvent data v3.2 model requirements via Open LCA v2.6. This study was originally designed with a scope limited to the contiguous United States in arid regions. Due to a lack of data availability, we have not limited our data to a specific geographic location and have used market average values. Additionally, this LCA is limited to a cradle-to-gate scope to align with the model and its capabilities. Any additional data found will be as recent as possible.

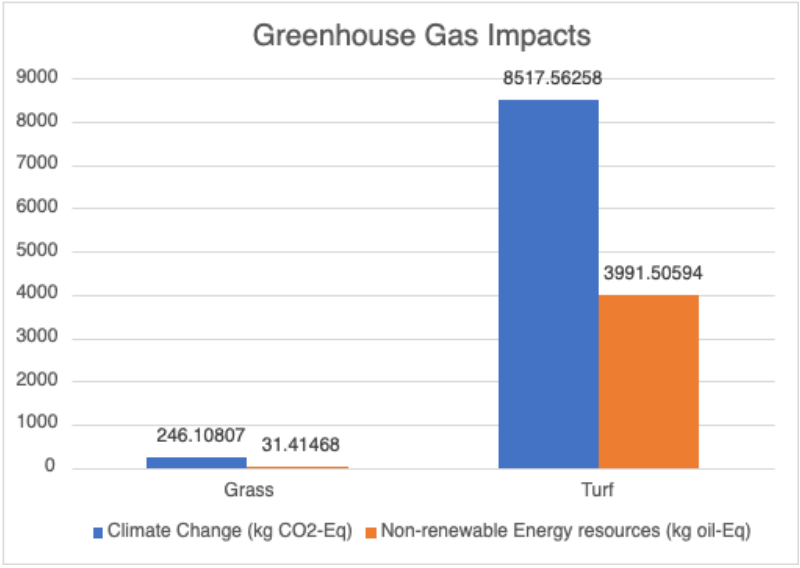
Environmental Impacts

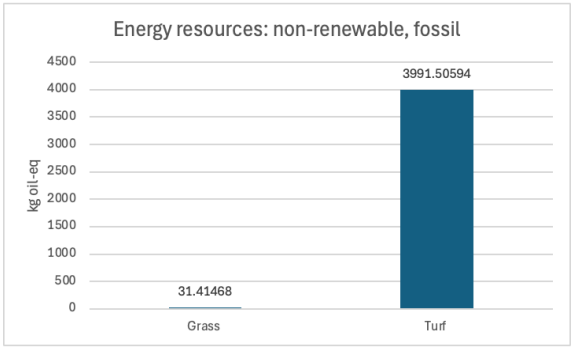
The impacts this LCA will be looking at include two categories: greenhouse gas impacts and water impacts. There are significant differences in greenhouse gas impacts, contributing to climate change and non-renewable energy resource depletion. Additionally, there are significant differences when looking at water impacts, including freshwater ecotoxicity, freshwater eutrophication, and water usage. Runoff in the environment can have significant ecosystem issues and can have a detrimental impact on overall ecosystem health. The ability to identify grass types with minimal ecosystem impact is crucial for reducing human-induced environmental stressors. With the housing industry continuously growing, we are at an impactful time to make these recommendations to homeowners.

3. Analysis

Results

To comparatively analyze the greenhouse gas and water impacts from grass and artificial turf, an OpenLCA impact analysis was completed. The results were imported into Excel for visualization. To assess the impacts of greenhouse gas emissions from grass and turf production, the categories "Energy Resources: non-renewable, fossil" and "Climate Change" were reviewed. Our results show a significant difference in outputs between grass and turf, with turf providing much larger greenhouse gas impacts. Regarding energy resources, the values for grass are less than 1% of the results for turf (figure 3). The Climate Change category showed similar results, with grass as 2.89% of the impacts from turf (figure 3).





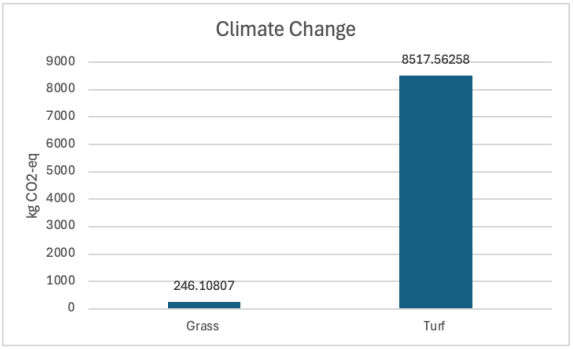
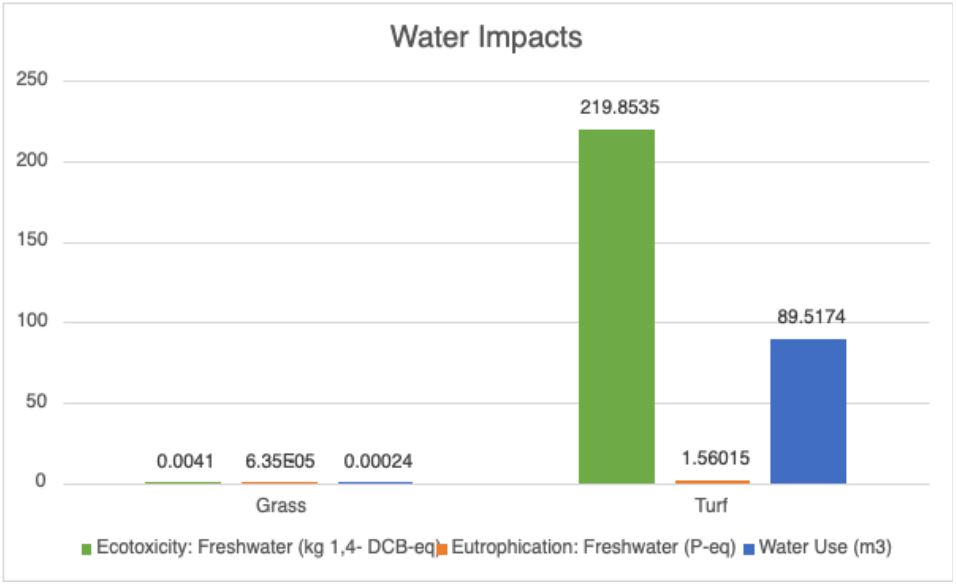


Figure 3. Comparison of results for non-renewable energy resources and climate change (greenhouse gas impacts) for grass and artificial turf.

To evaluate the impacts of water on turf and grass production, the distinctive categories "Ecotoxicity: Freshwater", "Eutrophication: Freshwater", and water use were evaluated (figure 4).



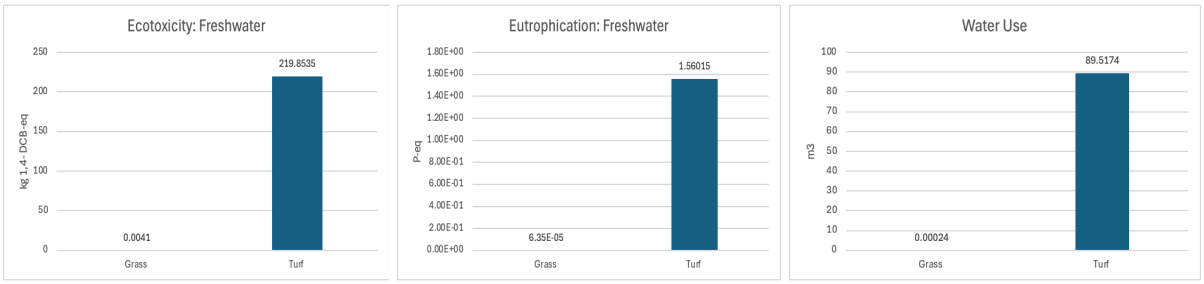
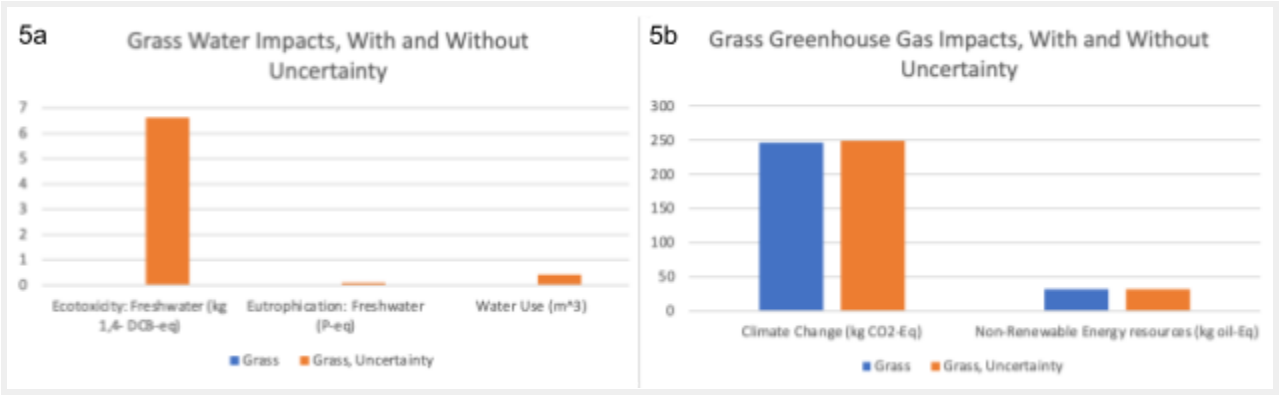


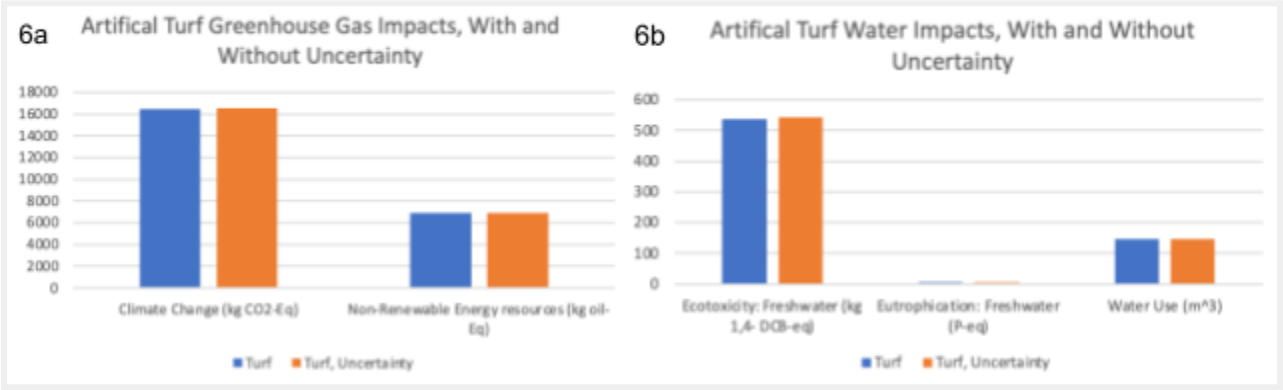
Figure 4. Water impacts from LCIA are shown in comparative analysis and broken down into distinctive categories.

Our results show a similar comparison to greenhouse gas impacts, where turf has a much more significant impact than grass. For all categories, the values determined for grass were less than 1% of the results from turf (figure 4).

When comparing model results with and without accounting for uncertainty, there is no significant difference in results for artificial turf. The same is true for natural grass in all but one category: freshwater ecotoxicity. Original model results calculated a 0.004 kg 1,4- DCB-eq ecotoxicity impact, whereas uncertainty-adjusted results output 6.604 kg 1,4- DCB-eq (figure 5a).



Figures 5a and 5b. Comparing uncertainty-adjusted impact model results for natural grass across the two primary categories.



Figures 6a and 6b. Comparing uncertainty-adjusted impact model results for artificial turf across the two primary categories.

Discussion

Our results showed a clear distinction in the impacts between grass and turf production. Turf production has a significantly higher impact on greenhouse gas emissions and water use than does grass. In every category, grass had less than 2.9% the impact than did turf, highlighting this clear distinction in product systems. We determined from our study that using grass (Kentucky Bluegrass) provides less of an environmental impact when assessing greenhouse gas and water impacts. It is recommended that homeowners consider planting natural grass on lawns rather than purchasing artificial turf.

The Sankey diagrams created from impact analysis visually highlight flow quantities for each product system. Artificial turf results show that carpet production is much more impactful than infill production (figure 7). This is likely due to the materials needed to create carpet being

primarily polymer-based. There are generally more upstream processes occurring to prepare the plastics for use in manufacturing. Additionally, crumb rubber represents (40% of infill mass and 36% of the total artificial turf mass, yet it has an extremely low environmental footprint (figure 7). Derived from used tires, it is classified as a waste product in the Ecoinvent database. This attributes all production emissions entirely to the original rubber and tire manufacturers. Natural grass is generally a more simple product, and potential environmental impacts are typically spread evenly across its cultivation and maintenance rather than being tied to a single, resource-heavy manufacturing process (figure 8).

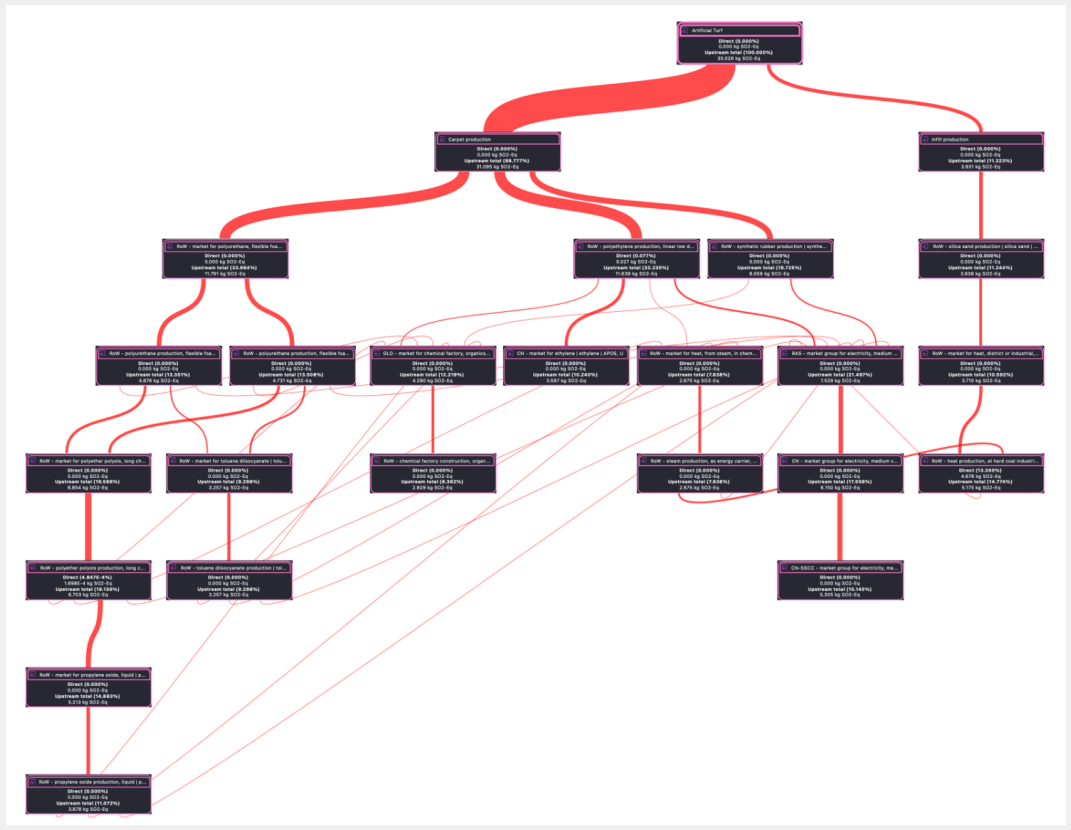


Figure 7. Artificial turf sankey diagram created with Ecoinvent data v3.2 via Open LCA v2.6 (Wernet et al., 2016).

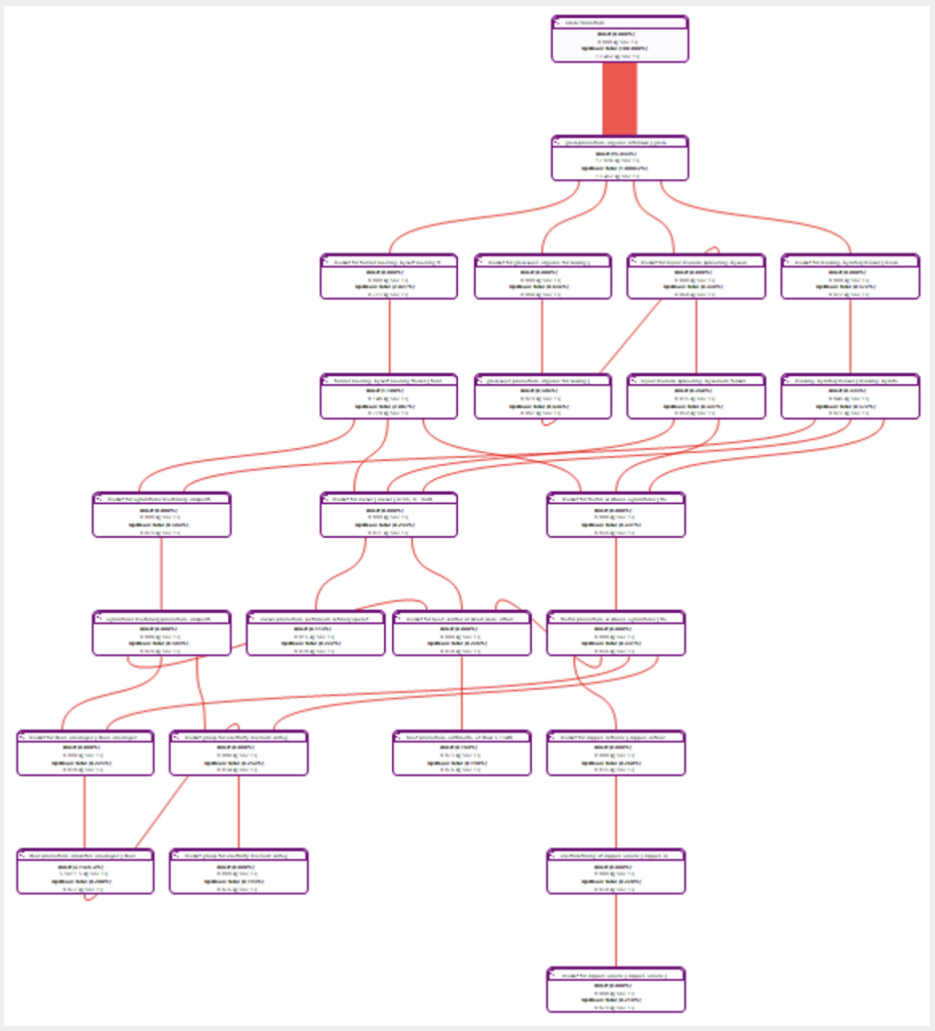


Figure 8. Natural grass sankey diagram created with Ecoinvent data v3.2 via Open LCA v2.6 (Wernet et al., 2016).

The largest potential source of error for our study was the creation of the turf production system in OpenLCA. Model input values were derived from an LCA by Falsafi et al., 2025, which assesses artificial turf in Finland using a functional unit of one standard FIFA-size football field (Falsafi et al., 2025). While the data in this study is measured directly and can be considered a primary source, it is not representative of the variability that would occur across countries. Incorporating non-specific data that can be applied to the entire world would address this uncertainty source. Additionally, we made various assumptions, as there were many types of artificial turf to choose from and production of turf varied from company to company.

If the use and end-of-life phases were included in this study, the results would change significantly. Grass requires significantly more water during the usage phase, where artificial turf only requires watering to clean when necessary. This would change the impacts of 'Water Use' and likely provide a more comprehensive perspective on the overall water usage of both systems. Additionally, the end-of-life phase for artificial turf is more intensive than natural grass, as there are more impacts to the disposal of artificial turf than the degradation of natural grass systems. These assumptions, as well as limitations of the cradle-to-gate study, would likely change the results, though we expect to see the same outcome regardless of these implications. Due to the nature of production and disposal of artificial turf, the results would likely stay the same, having a greater environmental impact than planting natural grass.

In comparison to our literature review, the findings of our LCIA showed some similarities and differences. Based on the literature, artificial turf was developed so homeowners would use less water during the use phase and reduce consumptive use. Regarding water use, the water impacts of production were not considered when making these statements. We believe that we would have found very different results regarding water use if we had included the use phase in this LCIA, which would have provided us with a comprehensive view of the water use of the entire system. Although we were not surprised to see that the literature aligned with the findings regarding the environmental impacts of turf, determining the LCIA; including the use phase, would provide us with a better recommendation for homeowners.

Conclusion

To conclude, the results of our LCIA analysis showed that turf has a much higher greenhouse gas and water impact during the production phase, which was expected. To provide a comprehensive outlook on total water use for both systems, the use phase and end-of-life phase should be considered, as grass uses much more water during the use phase. These results highlight the impact of including the production phase when comparing environmental impacts between natural grass and artificial turf for homeowners, but it doesn't show the whole picture.

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